Curriculum Vitae

Ananya Sinha

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Professional Summary: Currently working as a Junior Research Fellow at the Indian Institute of Technology, Roorkee, India. Interdisciplinary work interest, gained molecular, biochemical, structural, and computational insights in diagnosis and therapeutic targeting of disease-focused translational research. I have a strong biological background and passionate about integrating experimental and computational approaches to address complex biological questions.

Research Experience:

Position	Institute	Duration
1. Junior Research Fellow	**Indian Institute of Technology (IIT), Roorkee, India**	**July 2025 – Present**

Project title: Artificial Intelligence and Structure-based Approach to discover Broad Spectrum Antivirals against Emerging Viruses (Pan-alphaviruses & Pan-flaviviruses) funded by Defense Research and Development Organisation (DRDO)

Mentor: Prof. Pravindra Kumar

Investigating Zika virus capsid protein using integrated molecular and structural biology approaches to target RNA-binding sites for broad-spectrum antiviral design. Gained hands-on expertise in molecular cloning, protein expression and purification, Western blotting, site-directed mutagenesis, and enzyme activity assays. Complemented experimental work with molecular docking, simulations, and virtual screening, strengthening the ability to integrate experimental and computational methodologies.

2. Master's Thesis Work	**All India Institute of Medical Sciences (AIIMS), Delhi, India**	**May 2023 – Mar 2024**

1.Investigating the basis of susceptibility of individuals to microbial infections post-recovery from COVID-19 funded by Indian Council Medical Research (ICMR)

Mentor: Dr. Vikram Saini

I worked on host-pathogen interactions, focusing on the mechanisms underlying increased susceptibility to secondary microbial infections in COVID-19-recovered individuals. This study aimed to focus on the identification of key differentially regulated pathways to know the functional relevance of these gene expression deficits towards susceptibility to major nosocomial infections using molecular biology and infection biology techniques such as PBMCs Isolation, RNA Isolation, and cDNA Synthesis, Differential expression analysis, Cell Culture, and Infection Assay.

Research Internship Work:

Institute	Duration
1.All India Institute of Medical Sciences (AIIMS), Delhi, India	**Nov 2024 – Mar 2025**

Analysis of HE4 as a diagnostic and therapeutic response marker in epithelial ovarian carcinoma

Mentor: Prof. Amar Ranjan Singh

I validated HE4 protein levels (a biomarker for diagnosing ovarian cancer) in serum samples from suspected cancer patients using the Electrochemiluminescence technique (Cobas E411). HE4 and CA125 were tested on all samples to assess the diagnostic performance of HE4.

2.All India Institute of Medical Sciences (AIIMS), Delhi, India	**June 2022 – Aug 2022**

Investigation of biomarker expression patterns in ovarian cancer

Mentor: Prof. Amar Ranjan Singh

I analysed the expression of biomarkers such as WT1, Mucin-1, Mucin-4, and CA-125 in ovarian cancer using ELISA and real-time PCR. This expression analysis was considered to have a diagnostic role in ovarian cancer.

3.Indian Institute of Technology (IIT), Indore, India	June 2021– Aug 2021

PEG-PCL-based nanomedicines: A biodegradable drug delivery system and its application

Mentor: Prof. Chelvam Venkatesh

During the internship, I reviewed the research article on PEG-PCL-based nanomedicines and explored various nanoparticles, their synthesis methods, drug loading, delivery mechanisms, and ongoing advancements in polymer-based nanocarriers.

Education:

Year	Name of Degree	Board/University	Grade
2022-2024	M.Sc. in Biotechnology	All India Institute of Medical Sciences, Delhi, India	66.8% (Rank-6)
2019-2022	B.Sc. (Hons) Zoology	Banaras Hindu University, Varanasi India	8.1CGPA

Skills and Techniques:

Biochemistry: Recombinant protein expression and purification, Affinity purification using His-tagged proteins, Size Exclusion chromatography (AKTA Prime), protein expression analysis by native and SDS-PAGE, Western blotting, Enzyme kinetics analysis.

Molecular Biology: DNA and RNA Isolation, Plasmid Isolation, Molecular Cloning, Transformation, PCR, Real-time PCR, Site-Directed Mutagenesis, Gel Electrophoresis.

Bioinformatics: Pathway Analysis (Transcriptomics data) using KEGG and STRING, Virtual screening, docking, and validation using AutoDock (Vina and 4.2), PyMOL and molecular dynamics simulations (AlphaFold), Homology Modeling.

Cell Culture: Cell Culture (PBMCs, Bacterial Strain), Infection Assay (CFU), Plaque Assay.

Immunology: ELISA, Peripheral Blood Mononuclear Cells (PBMCs) Isolation.

Accomplishment: National-level entrance exams for pursuing academics:

- Advance Centre for Treatment, Research and Education in Cancer (ACTREC) JRF 2026 Qualified
- Graduate Aptitude Test in Engineering (GATE) 2025 Qualified: Biotechnology (Score-350), Life Sciences (Score-450)
- AIIMS Medical Biochemistry (Rank-02); AIIMS; M.Biotechnology (Rank-08)
- Graduate Aptitude Test of Biotechnology (GAT-B) 2022 (Rank-53)

Honors and Participation:

- **2025** Volunteered and participated in "International Conference on Advances in Biotechnology, Bioprocessing, and Structural Biology" jointly organized by Department of Biosciences and Bioengineering, IIT Roorkee, and The Biotech Research Society, India (BRSI).
- **2024** Winner in Quiz-o-con conducted in the symposium on "World Immunology Day" organized by the Department of Biochemistry, AIIMS New Delhi & the Indian Immunology Society.
- **2023** Second Prize Winner in "SCI-Q 2.0" (Scientific Quiz) organized by Society for Young Scientists (SYS), AIIMS, New Delhi.
- **2023** Participated in a poster presentation at the Max Cancer Congress, organized by the Max Institute of Cancer Care, on 29th & 30th of April 2023.
- **2019** Participated in the All India Essay Writing Event organized by Shri Ram Chandra Mission, United Nations Information Centre, and the Heartfulness Education Trust.

Membership, Leadership :

- Member of Society for Young Scientists (SYS), AIIMS, New Delhi.
- Member of National Service Scheme (NSS), Banaras Hindu University, India.
- Volunteer at Pledge A Smile Foundation "NOORI PROJECT" India.

COVER LETTER

When I first encountered biology in school, I was fascinated by how invisible molecular processes coordinate to drive cellular life. Over time, this fascination grew into a desire not only to understand biological systems, but also to observe how molecular components behave dynamically, respond to environmental hassle, and reorganize regulatory networks to shape cellular outcomes. This curiosity has guided my academic journey and is the reason I am strongly interested in pursuing doctoral research in the Topic: Changes to intron processing as a stress response.

My academic training has been interdisciplinary and centered on understanding how biological information flows from molecular regulation to cellular function. During my master's dissertation from All India Institute of Medical Sciences (AIIMS), New Delhi, I investigated RNA-level regulation and transcriptional control associated with host susceptibility to infection. This project provided me with hands-on experience in PBMCs isolation, RNA extraction, cDNA synthesis, cell culture (bacterial and PBMCs), infection assay with Class-II pathogen, transcriptomics data (differential pathway analysis) using KEGG and STRING. Through this work, I developed a strong foundation in post-transcriptional regulation and gained insight into how gene regulatory networks dynamically shape cellular phenotypes under stress and infection conditions.

Currently, as a Junior Research Fellow at the Indian Institute of Technology (IIT), Roorkee, India, I investigate the zika virus capsid protein, employing an interdisciplinary experimental and computational approach to target RNA-binding sites for antiviral discovery. This project has provided me with hands-on experience in recombinant protein techniques (cloning, expression, purification, mutagenesis) and computational methods (docking, MD simulations, virtual screening). In addition, my internship in a cancer biology laboratory, where I worked on biomarker expression analysis in ovarian cancer patients using ELISA, qPCR, and electrochemiluminescence platforms, allowed me to connect molecular regulatory changes with disease outcomes. This further reinforced my interest in research that bridges fundamental regulatory biology with physiological and pathological relevance.

I am particularly interested in work investigating stress-induced changes in RNA processing and the role of introns in cellular adaptation and chromosome instability. The idea that RNA splicing, transcript processing, and intron regulation function not merely as housekeeping mechanisms but as dynamic regulatory systems driving stress responses represents a powerful and conceptually exciting framework. I am especially motivated to explore how stress alters RNA processing pathways, how intron-containing genes are selectively regulated, and how these post-transcriptional programs contribute to cellular plasticity and adaptation.

I am highly motivated to train in RNA biology, post-transcriptional regulation, and stress-responsive gene control as well as chance to learn new techniques. I feel I am ready to start a new journey guided by your expertise. I would be glad to discuss how my experience and research interests align with your laboratory's goals.

Thank you for considering my application.

Sincerely,
Ananya Sinha.